UT Regional 2023



Dynamic Planet

Division B/C

Notes: CLASS COPY (DO NOT WRITE)

Student Names: _____

_____

Team Name: _____

Team Number: B/C___

Stream Drainage Systems

1. Which of these is a Strahler 12th order stream? (2)
    a. Nile
    b. Mississippi
    c. Amazon
    d. Yangtze

2. What is the most common drainage pattern? (2)
    a. Dendritic
    b. Radial
    c. Trellis
    d. Rectangular

3. What percent of water is found within lakes, streams, groundwater, and the river? (2)
    a. 5%
    b. 2%
    c. 7%
    d. 10%

4. Label river A and its tributaries on the answer sheet using the classical stream order. (4)

5. Label river B and its tributaries on the answer sheet using the Strahler stream order. (4)

6. Label river C and its tributaries on the answer sheet using the Shreve stream order. (4)

7. Describe how the dendritic drainage pattern forms. (3)

8. Describe how the trellis drainage pattern forms. (3)

River Sediment

9. Which of the following characteristics would give the fastest settling sediment? (2)
    a. Flat, Dense
    b. Spherical, Dense
    c. Flat, Less Dense
    d. Spherical, Less Dense

10. What does it mean when a trunk stream is in balance? (2)
    a. The trunk has the same amount of tributaries on both sides
    b. The rate of erosion equals the rate of deposition
    c. The rate of water inflow is the same as outflow
    d. The trunk has symmetrical depth on both sides

11. If a stream's velocity increases by two, by what factor does its competence increase? (2)
    a. 1
    b. 2
    c. 4
    d. 8

12. What two factors control the type and amount of material carried in suspension? (2)
    a. Water Velocity, Settling Velocity
    b. Discharge Rate, River Width
    c. Water pH, Water Temperature
    d. Water Capacity, Water Depth

13. Which of the following loads is least affected by the stream velocity? (2)
    a. Suspended load
    b. Dissolved Load
    c. Bedload
    d. None of the above

14. How does a pothole form in a river? (3)

15. Describe what the dissolved load is: (3)

16. Describe what the suspended load is: (3)

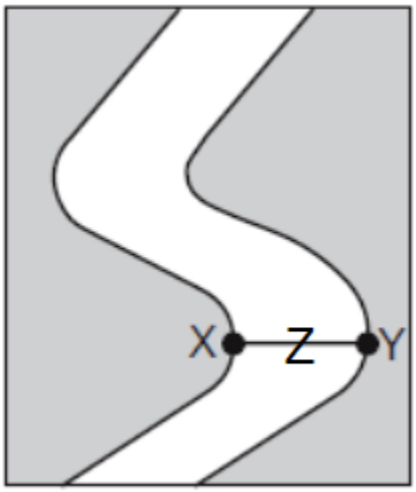
17. Describe the process of saltation in the bedload. (3)

18. When does the Dissolved load leave the water (aka stop being transported)? (2)

19. When does the Suspended load leave the water (aka stop being transported)? (2)

20. What is the difference between competence and capacity? (3)

Stream Channels and Measurement



21. Where is the maximum velocity in the above meandering stream? (2)
    a. X
    b. Y
    c. Z

22. Where is the deepest point in a meandering stream? (2)
    a. X
    b. Y
    c. Z

23. What's it called when erosion cuts through between two meanders and isolates a smaller bend? (2)
    a. Oxbow Lake
    b. Orphan Lake
    c. Fluvial Lake
    d. Tarn

24. What is one benefit of ADCP over normal current meters? (2)
    a. ADCP is cheaper than normal current meters
    b. ADCP creates profiles for the majority of the cross section instead of just at point locations
    c. ADCP is less intrusive to the wildlife
    d. ADCP allows for measurement of river under an ice cover



25. What is shown in the image above? (2)
    a. Pygmy meter
    b. Vortex Meter
    c. Flow Probe
    d. Current Meter

26. Why are braided channels wider and shallower than meandering streams? (3)

27. What about the environment determines whether a stream will meander or be braided? (3)

28. How does an Acoustic Doppler Current Profiler measure the velocity of water? (3)

29. Assume you have a river flowing with a velocity of 15 m/s. This river is in the shape of a semicircle with a radius of 3 meters. Calculate the discharge of this river (show your work!) (3)

30. Why does a river flow the quickest near the center of the stream? (3)

Groundwater

31. Which of the following has the greatest porosity? (2)
    a. Gravel
    b. Clay
    c. Sand
    d. Igneous Rocks

32. Which of the following would be best for an aquitard? (2)
    e. Gravel
    f. Clay
    g. Sand
    h. Igneous Rocks

33. How can a highly porous rock have poor permeability? (3)

34. How do drawdown and cones of depression lead to the depletion of nearby wells? (3)

35. In sedimentary rocks, most porosity is provided by space between the individual grains. However, how is porosity provided in more tightly interlocked rock types such as igneous and metamorphic rocks? (3)

36. When considering the path of groundwater underground, explain how water can sometimes turn upwards and travel against the force of gravity? (3)

37. Why are urban areas more susceptible to flash floods than more natural areas? (3)

38. Usually, a well must be below the water table in order to contain water; however, this is not the case for artesian wells. Why is this? (3)

Lakes and Karst Topography

39. Which of the following is not a common rock type found in karst topographies? (2)
    a. Limestone
    b. Dolomite
    c. Gypsum
    d. Quartz

40. Which acid is most commonly the cause of karst topography? (2)
    a. Sulfuric Acid
    b. Carbonic Acid
    c. Hydrochloric Acid
    d. Nitric Acid

41. Why might water from wells in karst topography be unsafe to drink from? (3)

42. How do speleothems form within cave systems? (3)

43. In which season is the epilimnion most distinct? (2)
    a. Summer
    b. Spring
    c. Fall
    d. Winter

44. When lakes are heavily stratified, which resource is the hypolimnion most limited in? (2)
    a. Phosphorus
    b. Nitrogen
    c. Oxygen
    d. Carbon

45. Briefly describe how a kettle lake forms. (3)

46. Tectonic motion may lead to the creation of a depression in the land called a graben. Describe how this graben may then become a lake. (3)

47. How can glacial moraines create lakes? (3)

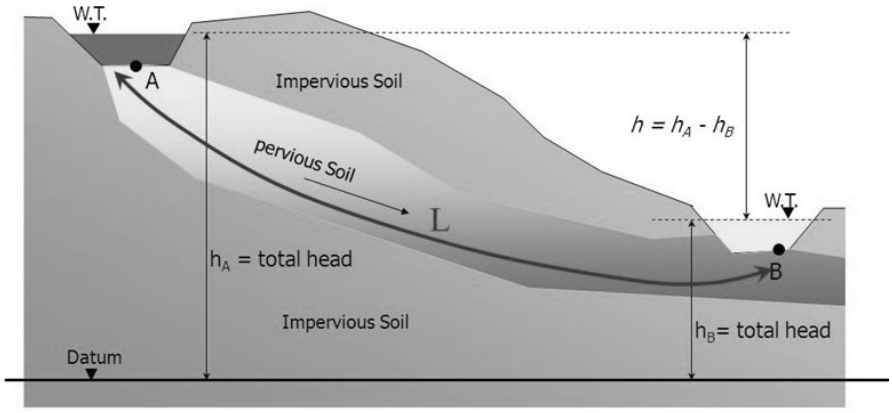
Land Use Changes

48. Why do flood-control dams erode deltas and floodplains? (3)

49. Why does dam effectiveness decrease over time? (3)

50. How do artificial cutoffs reduce flood damage? (3)

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51. Calculate the specific flow rate of the above water system. The distance between A and B is 9 kilometers. A is at a height of +1000 meters and B is at a height of +200 meters. The hydraulic conductivity is 12.5 meter/day. (5)